

Fine-Grained Sentiment Analysis for Product Feedback Analysis

Jason Widjaja¹, Cheng Zeng¹, Hamzah Alamdeen¹, Mingyang Cai¹, and Bryan Che-Sheng Jen¹

¹UNSW Sydney, Australia, z5494973@unsw.edu.au, z5431483@unsw.edu.au, z5428209@unsw.edu.au, z5402186@unsw.edu.au, z5360176@unsw.edu.au

ABSTRACT

Aspect-based sentiment analysis (ABSA) provides finer-grained insight into customer feedback by predicting sentiment toward specific aspects mentioned in a review rather than assigning a single label to the entire text. In this paper, we study aspect level sentiment classification for product and service feedback in the restaurant and laptop domains. We use the SemEval-2014 ABSA benchmark together with a custom Yelp-based restaurant dataset annotated for explicit aspect terms and three sentiment labels with each being positive, negative, and neutral.

We compare four approaches that span classical machine learning (TF-IDF with Logistic Regression), encoder-based transformers (fine-tuned BERT), generative language models (LoRA-based supervised fine-tuning of Qwen2.5-1.5B-Instruct), and prompt-based reasoning (zero-shot Chain-of-Thought prompting with GPT-5.4-mini). Experiments evaluate both in-domain performance and cross-domain generalisation by training on restaurant data alone or on combined restaurant and laptop data. They are then tested on restaurant and laptop benchmarks.

Results show clear differences across methods. TF-IDF with Logistic Regression provides a simple and interpretable baseline but is less robust under multi-domain training. Fine-tuned BERT achieves strong performance after supervision, while zero-shot BERT performs poorly. Qwen with LoRA adaptation shows strong gains after fine-tuning, especially on the laptop dataset. GPT-5.4-mini with prompting performs competitively in zero-shot settings, but Chain-of-Thought prompting yields only limited improvement over direct prompting. Overall, our findings highlight the importance of supervised adaptation for ABSA and show that larger instruction-tuned models and transformer encoders can substantially outperform traditional baselines while cross-domain gains depend on both model architecture and training strategy.

PROJECT SCOPE

Part	Item	Credits	Project details
A	NLP Problem	5	Aspect-Based Sentiment Analysis (ABSA), specifically aspect-level sentiment polarity classification given a sentence and a target aspect term.
A	Text Source / Domain	5	Restaurant and laptop review text. Data comes from the SemEval-2014 ABSA benchmarks and a custom Yelp restaurant dataset annotated for explicit aspect terms and polarity.
A	Part A subtotal	10	
B	Use two existing datasets	10	SemEval-2014 Restaurant and SemEval-2014 Laptop datasets.
B	Create your own labelled dataset	20	Custom Yelp restaurant dataset annotated for explicit aspect terms and three sentiment labels: positive, negative, and neutral.
B	Use an existing lexicon	0	Not used in this project.
B	Part B subtotal	30	
C	Implement a rule-based or statistical model as a baseline	Essential	TF-IDF + Logistic Regression baseline for aspect-level sentiment classification.
C	Use an existing pre-trained / fine-tuned model	5	Zero-shot BERT baseline.
C	Use an existing pre-trained / fine-tuned model	5	Zero-shot Qwen2.5-1.5B-Instruct baseline.
C	Use an existing pre-trained / fine-tuned model	5	GPT-5.4-mini prompt-based baseline.
C	Fine-tune a model based on a dataset	20	Fine-tuned BERT for three-class aspect-level sentiment classification.
C	Fine-tune a model based on a dataset	20	LoRA-based supervised fine-tuning of Qwen2.5-1.5B-Instruct.
C	Extend a method	30	Few-shot Chain-of-Thought prompting for aspect-aware sentiment reasoning with GPT-5.4-mini.
C	Integrate a language model with external tools	0	Not used in this project.
C	Part C subtotal	85	
D	Quantitative Evaluation	10	Accuracy, macro F1, per-class precision/recall/F1, confusion matrices, and cross-domain evaluation across restaurant and laptop datasets.
D	Qualitative Evaluation	5	Misclassification analysis with representative error cases in both restaurant and laptop domains.
D	Demo	10	Gradio demo for interactive testing of aspect-level sentiment prediction.
D	Part D subtotal	25	
Total claimed credits		150	

1 INTRODUCTION

Online review platforms such as Yelp and Google Reviews contain large amounts of user-generated feedback about products and services. While overall star ratings provide a coarse summary of customer satisfaction, they do not reveal how users feel about specific aspects of an item or service. For example, a restaurant review may praise the food but criticise the service, and this mixed sentiment is lost when the review is reduced to a single overall label.

Aspect-Based Sentiment Analysis (ABSA) addresses this limitation by identifying the sentiment expressed toward a specific target aspect within a sentence. This makes review analysis more informative for both businesses and customers, since it supports fine-grained understanding of what is liked or disliked. In this project, we study aspect-level sentiment classification in the restaurant and laptop domains using both benchmark and custom-annotated data.

We compare four approaches that span classical machine learning, supervised transformer fine-tuning, parameter-efficient adaptation of an instruction-tuned language model, and prompt-based reasoning. Our experiments evaluate both in-domain performance and cross-domain generalisation. Overall, the results show that supervised contextual and generative models outperform the classical baseline, while the neutral class remains the most difficult label across domains.

2 RELATED WORK

Aspect-Based Sentiment Analysis (ABSA) extends conventional sentiment classification by predicting sentiment toward a specific target aspect rather than assigning a single label to an entire sentence. The task was standardised by SemEval-2014 Task 4, which introduced benchmark datasets and evaluation settings for aspect term extraction and aspect-level polarity classification in the restaurant and laptop domains (Pontiki et al., 2014). These datasets remain widely used benchmarks for both in-domain and cross-domain ABSA evaluation.

A common traditional baseline for text classification is TF-IDF with a linear classifier such as Logistic Regression. TF-IDF represents text using weighted lexical features, while Logistic Regression provides a simple and efficient model for sparse inputs (Salton and Buckley, 1988). Although this approach does not explicitly model the interaction between a target aspect and its context, it remains a useful reference point for comparison with stronger neural methods.

Neural ABSA models improved on such baselines by explicitly conditioning prediction on the target aspect. Earlier attention-based and memory-based approaches showed that different parts of the same sentence may express different sentiments toward different aspects, making target-aware contextual modelling important for aspect-level classification (Wang et al., 2016; Tang et al., 2016). Pretrained transformer encoders such as BERT later achieved further gains through stronger contextual representations and became a standard supervised baseline for ABSA (Devlin et al., 2019; Xu et al., 2019).

More recently, instruction-tuned large language models have been applied to classification by framing the task as constrained text generation. In this setting, the model receives a review and a target aspect, then generates a polarity label. We follow this paradigm using Qwen2.5-1.5B-Instruct and adapt it with LoRA, a parameter-efficient fine-tuning method that updates only a small number of trainable weights while keeping the pretrained backbone frozen (Yang et al., 2025; Qwen Team, 2024; Hu et al., 2021).

Prompting-based methods have also become important for evaluating LLM reasoning ability. Chain-of-Thought (CoT) prompting can improve performance by encouraging the model to generate intermediate reasoning steps before producing an answer (Wei et al., 2022). In this work, we compare direct prompting and few-shot CoT prompting with GPT-5.4-mini, allowing us to contrast supervised adaptation with inference-time prompt engineering.

3 DATASET

We use two datasets for aspect-level sentiment classification: a restaurant dataset and a laptop dataset. The restaurant dataset combines the SemEval-2014 restaurant benchmark with a custom Yelp-based dataset annotated in this project, while the laptop dataset is the SemEval-2014 laptop benchmark (Pontiki et al., 2014). The SemEval restaurant reviews were originally adapted from the Citysearch restaurant review corpus of Ganu et al. (2009). This setup allows us to evaluate both in-domain performance and cross-domain generalisation.

3.1 Data Sources and Annotation Scope

The custom Yelp dataset is derived from the Yelp Open Dataset (Yelp Inc., 2024). We restrict our analysis to the *tip* portion of the dataset, which consists of short review-like texts that are suitable for aspect-based sentiment analysis. For this custom dataset, we annotate only *explicitly mentioned* aspect terms and assign one of three polarity labels: *positive*, *negative*, or *neutral*. Examples containing only implicit aspect categories are excluded, and the *conflict* label is removed because of its very low frequency.

Restricting annotation to explicit targets makes the aspect-level sentiment task more clearly defined and improves consistency across annotators. However, this also reduces coverage, since naturally occurring implicit opinions such as “Expensive but delicious” are not retained unless an explicit target is present.

3.2 Annotation Reliability

The Yelp subset was annotated by two annotation groups following guidelines inspired by the SemEval-2014 ABSA task. Inter-annotator agreement was assessed on an overlapping subset using Cohen’s Kappa. As shown in Table 2, the two groups achieved agreement scores of 0.83 and 0.89, indicating generally strong annotation consistency.

Group	Metric	Score
Group 1 annotation	Cohen’s Kappa	0.83
Group 2 annotation	Cohen’s Kappa	0.89

Table 2. Inter-annotator agreement on the annotated Yelp subset.

3.3 Dataset Summary

Table 3 summarises the final datasets used in this study. The restaurant dataset contains the combined SemEval-2014 restaurant and custom Yelp annotations, while the laptop dataset consists of the SemEval-2014 laptop benchmark only. The restaurant dataset is larger and more skewed toward the positive class, while the laptop dataset is smaller but more balanced between positive and negative sentiment. Additional annotation details and exploratory data analysis are provided in Appendix A.

4 METHODS

4.1 Aspect-Level Classification Setup

All models in this study are used for aspect-level sentiment polarity classification. Each instance consists of a review sentence and a target aspect term, with one polarity label assigned per aspect. If a sentence contains multiple aspect terms, it appears as multiple instances in the dataset. The task is formulated as a three-class classification problem with labels *positive*, *negative*, and *neutral*.

We report accuracy, macro F1, and per-class precision, recall, and F1-score. Macro F1 is treated as the main comparison metric because the class distribution is imbalanced and macro F1 gives equal weight to each class. We also use confusion matrices and model-specific interpretability analysis where appropriate.

To evaluate both in-domain performance and cross-domain generalisation, we use four train–test settings: restaurant-trained models tested on restaurant data, restaurant-trained models tested on laptop data, restau-

Statistic	Restaurant dataset	Laptop dataset
Number of samples	3845	1462
Number of annotated aspect terms	6912	2313
Positive instances	4311	987
Negative instances	1322	866
Neutral instances	1279	460
Conflict instances removed	98	45

Table 3. Summary statistics of the datasets used in this study. The restaurant dataset combines SemEval-2014 restaurant reviews with the custom Yelp dataset, while the laptop dataset is SemEval-2014 laptop.

rant+laptop trained models tested on restaurant data, and restaurant+laptop trained models tested on laptop data.

4.2 TF-IDF + Logistic Regression

As a traditional machine learning baseline, we use TF-IDF features with a Logistic Regression classifier. For each example, the target aspect is marked in the sentence using special tokens so that the model can distinguish sentiment toward the target aspect from sentiment toward the sentence as a whole. The resulting aspect-marked text is converted into sparse lexical features and used to train a class-weighted Logistic Regression model.

We train two versions of this baseline: one on the restaurant training split only, and one on the combined restaurant and laptop training data. In addition to quantitative evaluation, we analyse model behaviour using feature weights and SHAP-based explanations.

4.3 BERT Base Uncased

We use `bert-base-uncased` as a strong supervised transformer baseline. The input is represented as the sentence together with the target aspect term, allowing the model to condition its prediction on both the review context and the specified aspect. A classification layer is applied on top of the final `[CLS]` representation to predict the three sentiment classes.

We evaluate both a zero-shot variant and a fine-tuned variant. For supervised training, the full model is updated on the labelled training data using class-weighted loss to reduce the impact of label imbalance. Full implementation details are provided in Appendix B.

4.4 Qwen2.5-1.5B-Instruct with LoRA

To evaluate instruction-tuned generative modelling for ABSA, we adapt `Qwen2.5-1.5B-Instruct` using LoRA. The task is framed as constrained text generation: given a review and a target aspect, the model is prompted to generate one of the three polarity labels. LoRA allows efficient task adaptation by training only a small set of additional parameters while keeping the pretrained backbone fixed.

We train the model using supervised fine-tuning with class reweighting to better handle the minority classes. We consider both a restaurant-only LoRA model and a sequential setting in which the model is further adapted using laptop data. The full prompt format, weighted loss description, and hyperparameters are given in Appendix B and Appendix F.

4.5 Prompting with GPT-5.4-mini

We also evaluate prompt-based classification using GPT-5.4-mini. Two prompting strategies are compared: a direct zero-shot prompt that asks the model to predict the sentiment label for a target aspect, and a few-shot Chain-of-Thought (CoT) prompt that encourages the model to reason through the aspect mention and its local context before producing the final polarity.

This comparison allows us to contrast supervised adaptation against inference-time prompt engineering. GPT-5.4-mini is used separately on the restaurant and laptop test sets, and the resulting predictions are evaluated using the same metrics as the other models. The exact prompt templates are provided in Appendix B.

5 RESULTS

5.1 Restaurant Results

Table 4 reports performance on the combined restaurant dataset. The majority baseline performs poorly under Macro F1, showing that the task cannot be solved by exploiting class imbalance alone. TF-IDF + Logistic Regression provides a competitive classical baseline, but it remains clearly below the stronger contextual and generative models.

BERT performs very poorly in the zero-shot setting (Macro F1 = 0.13), showing that pretrained representations alone are insufficient for this task without supervised adaptation. Fine-tuning improves BERT substantially to 0.73–0.74 Macro F1. Qwen in the zero-shot setting already performs reasonably well on positive and negative classes, but weak neutral performance limits its overall Macro F1. LoRA adaptation improves neutral predictions and raises Macro F1 to 0.74.

GPT-5.4-mini achieves the strongest zero-shot result on this dataset (Macro F1 = 0.77). However, adding Chain-of-Thought prompting does not improve performance further, suggesting that prompt-based reasoning offers limited benefit once the model already performs strongly on this domain.

Table 4. Per-class F1 and Macro F1 on the combined restaurant dataset.

Model	Positive	Negative	Neutral	Macro F1
Majority baseline	0.77	0.00	0.00	0.26
TF-IDF + LR (rest.)	0.80	0.53	0.50	0.61
TF-IDF + LR (rest. + laptop)	0.81	0.56	0.51	0.63
BERT (zero-shot)	0.11	0.29	0.00	0.13
BERT (rest.)	0.89	0.74	0.56	0.73
BERT (rest. + laptop)	0.90	0.74	0.59	0.74
Qwen (zero-shot)	0.87	0.79	0.17	0.61
Qwen-LoRA (rest.)	0.82	0.80	0.52	0.71
Qwen-LoRA (rest. + laptop)	0.88	0.79	0.55	0.74
GPT-5.4-mini (zero-shot)	0.91	0.82	0.56	0.77
GPT-5.4-mini (CoT)	0.91	0.82	0.54	0.76

5.2 Laptop Results

Table 5 reports performance on the laptop dataset. The overall pattern is similar, but the relative gains from domain adaptation differ across models. Zero-shot BERT again performs poorly, while fine-tuning on restaurant data transfers well to the laptop domain and raises Macro F1 to 0.79. Additional laptop fine-tuning changes the class balance of predictions, but does not improve the overall Macro F1.

Qwen benefits the most from additional domain-specific supervision. Its zero-shot Macro F1 is 0.69, which improves to 0.74 after restaurant-only LoRA and to 0.86 after further adaptation on laptop data. This is the best result on the laptop dataset and suggests that Qwen benefits strongly from sequential domain adaptation.

GPT-5.4-mini also performs strongly in the zero-shot setting (Macro F1 = 0.84), but Chain-of-Thought prompting again yields little improvement. Across both datasets, the neutral class remains the hardest to predict, and multi-domain training helps Qwen more than it helps TF-IDF or BERT.

Table 5. Per-class F1 and Macro F1 on the laptop dataset.

Model	Positive	Negative	Neutral	Macro F1
Majority baseline	0.62	0.00	0.00	0.21
TF-IDF + LR (rest.)	0.76	0.70	0.53	0.66
TF-IDF + LR (rest. + laptop)	0.71	0.71	0.46	0.63
BERT (zero-shot)	0.58	0.00	0.06	0.21
BERT (rest.)	0.83	0.85	0.69	0.79
BERT (rest. + laptop)	0.85	0.65	0.83	0.79
Qwen (zero-shot)	0.84	0.85	0.37	0.69
Qwen-LoRA (rest.)	0.78	0.86	0.58	0.74
Qwen-LoRA (rest. + laptop)	0.90	0.91	0.75	0.86
GPT-5.4-mini (zero-shot)	0.93	0.90	0.68	0.84
GPT-5.4-mini (CoT)	0.92	0.89	0.70	0.83

6 DISCUSSION

6.1 Cross-Domain Error Patterns

The results show that performance differences across models are driven not only by overall accuracy, but also by the type of reasoning required for aspect-level sentiment classification. Across both domains, the neutral class remains the most difficult to predict. This is partly due to class imbalance, but also reflects deeper ambiguity in the task: neutral cases are often weakly expressed, context-dependent, or close to mildly positive and mildly negative examples.

The restaurant and laptop domains also expose different failure modes. In restaurant reviews, errors often arise from implicit sentiment, contrastive clauses, aspect-level attribution, and figurative language. In laptop reviews, errors more often involve conditional reasoning, technical phrasing, and numerical comparisons. These patterns suggest that domain transfer is not only a matter of vocabulary shift, but also a shift in the type of linguistic reasoning required.

Overall, BERT is effective after supervised fine-tuning but tends to rely heavily on local lexical cues, which makes it more brittle on contrastive, conditional, and domain-specific examples. Qwen and GPT-5.4-mini show stronger compositional and contextual reasoning, especially on technical and implicitly structured cases, although all models continue to struggle on more difficult examples such as counterfactuals, irony, and sentiment requiring discourse-level interpretation. Representative examples and additional qualitative analysis are provided in Appendix D.

6.2 Model Robustness and Limitations

Table 6 summarises the main linguistic challenges observed across models. The pattern is consistent with the quantitative results: classical and encoder-based approaches are more sensitive to surface lexical cues, while larger generative models are relatively better at handling compositional sentiment and domain-specific phrasing. However, no model is robust across all challenge types.

These findings highlight several limitations of current ABSA systems. First, explicit aspect-level classification still struggles with naturally ambiguous or weakly expressed sentiment, especially for the neutral class. Second, cross-domain transfer remains incomplete, since models that perform well on restaurant reviews may still fail on technical laptop descriptions. Third, even strong models remain unreliable on examples requiring pragmatic or discourse-level inference. Future work could improve robustness by expanding annotation coverage for implicit opinions, improving neutral-class consistency, and incorporating training objectives that better capture compositional and context-dependent sentiment.

Table 6. Summary of model robustness across key linguistic challenges in ABSA. Legend: ✓ = handles reliably, △ = partial / inconsistent, ✗ = struggles.

Challenge	Example	ML	BERT	Qwen	GPT
Double negation (polarity)	“not bad”	△	△	✓	✓
Contrastive sentiment (composition)	“expensive but tasty”	✗	✗	✓	✓
Conditional reasoning (counterfactual)	“would be perfect if . . .”	✗	✗	△	△
Domain-specific sentiment (technical phrasing)	“drivers updated ok”	✗	✗	✗	✓
Figurative language (non-literal sentiment)	irony / understatement	✗	✗	✗	△

7 CONCLUSION

This paper compared four approaches for aspect-based sentiment analysis across restaurant and laptop reviews: TF-IDF with Logistic Regression, fine-tuned BERT, LoRA-adapted Qwen2.5-1.5B-Instruct, and prompt-based classification with GPT-5.4-mini. Using SemEval-2014 benchmarks together with a custom Yelp restaurant dataset, we evaluated both in-domain performance and cross-domain generalisation.

Overall, the results show that supervised contextual and generative models outperform the classical TF-IDF baseline on aspect-level sentiment classification. Fine-tuned BERT provides a strong supervised encoder baseline, while Qwen with LoRA achieves the strongest gains from sequential domain adaptation, especially on the laptop dataset. GPT-5.4-mini performs competitively in the zero-shot setting, but Chain-of-Thought prompting provides little additional benefit over direct prompting.

Across all settings, the neutral class remains the hardest to predict, reflecting both label imbalance and the difficulty of weakly expressed or context-dependent sentiment. The qualitative analysis further shows that current models still struggle with implicit sentiment, contrastive clauses, conditional reasoning, technical phrasing, and discourse-level interpretation.

Future work should focus on improving annotation consistency for neutral cases, extending the dataset to include implicit aspects, and developing methods that better capture compositional and context-dependent sentiment across domains.

A ANNOTATION DETAILS AND EXPLORATORY DATA ANALYSIS

A.1 Custom Yelp Annotation Procedure

Our annotation design was inspired by the SemEval-2014 ABSA task, particularly the subtasks of aspect term extraction and aspect polarity classification. However, our dataset focuses only on *explicitly mentioned* aspect terms and their associated sentiment polarity.

To create the dataset, a randomly selected subset of Yelp tips was annotated by two groups, each consisting of two student annotators. One group annotated the first half of the selected samples, while the second group annotated the second half. Inter-annotator agreement was computed on an overlapping subset.

A.2 Explicit Aspect Terms

Similar to SemEval-2014, we annotate only explicit aspect terms that are directly stated in the text. For example, in the sentence “The battery life is terrible but the screen is excellent”, the aspect terms are *battery life* and *screen*. Restricting the dataset to explicit targets reduces ambiguity during annotation and makes the polarity classification task more clearly defined.

However, this choice excludes cases where the sentiment target is only implied. For example, in “Expensive but delicious”, the words *expensive* and *delicious* suggest the categories of price and food respectively, but no explicit aspect term is mentioned. Since no explicit target can be extracted, such examples are discarded from our dataset. This improves annotation consistency, but may reduce coverage of naturally occurring opinion expressions.

A.3 Polarity Labels

For each extracted aspect term, we assign one of three polarity labels: *positive*, *negative*, or *neutral*. We initially also considered the *conflict* label, following SemEval-2014. However, conflict cases were extremely rare in our annotated subset, so they were removed from the final dataset to avoid severe class imbalance and unstable learning behaviour.

While this simplifies the classification task, it may also limit the model’s ability to represent genuinely mixed sentiment toward a single aspect. This trade-off between label coverage and class balance is an important limitation of the current dataset design.

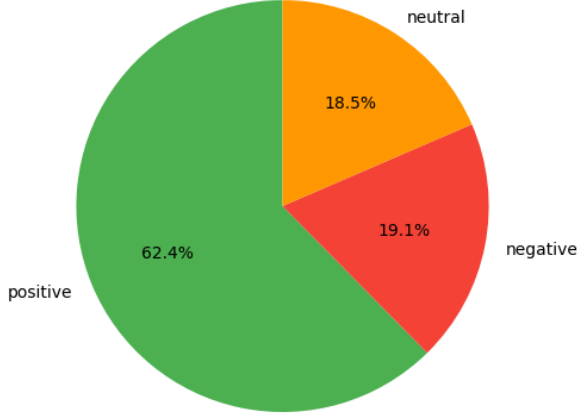
A.4 Challenges of Annotation

Aspect-based annotation is inherently challenging. Annotators must first determine whether a sentence contains an explicit aspect term, then identify the exact span of that term, and finally assign the correct sentiment polarity with respect to that target. Disagreements may arise due to boundary selection, ambiguity in sentiment strength, or uncertainty about whether a target is explicit or implicit.

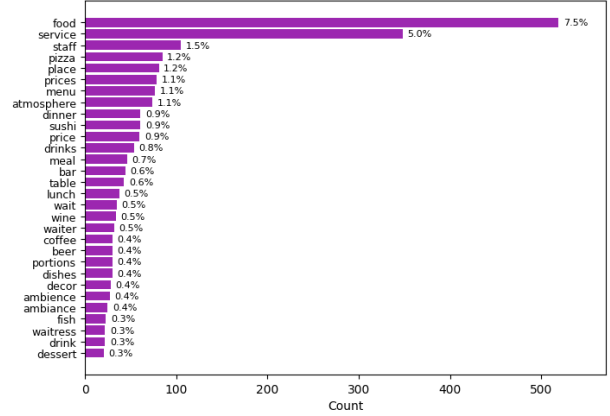
These challenges are particularly pronounced in short review texts such as Yelp tips, where the context is limited and sentiment may be expressed informally, indirectly, or through slang. As a result, even carefully designed guidelines may not eliminate all annotation ambiguity.

A.5 Exploratory Data Analysis

Figure 1 and Figure 2 show the polarity distribution and most frequent aspect terms in the restaurant and laptop datasets respectively. These figures complement the summary statistics in Table 3 by illustrating both class imbalance and the different lexical profiles of the two domains.

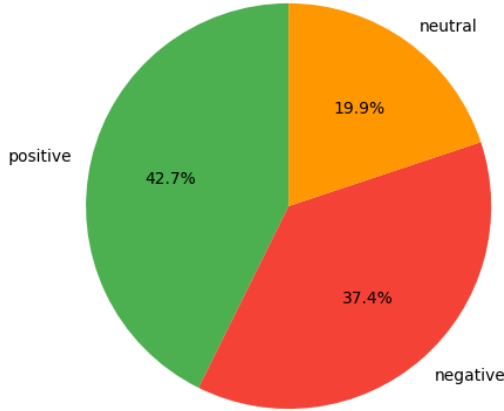


(a) Polarity distribution of aspect annotations.

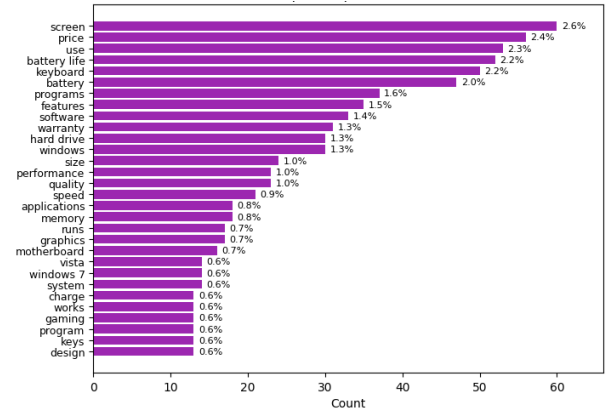


(b) Top 30 most frequent aspect terms.

Figure 1. Exploratory data analysis for the restaurant dataset, which combines SemEval-2014 restaurant reviews with the custom Yelp dataset.



(a) Polarity distribution of aspect annotations.



(b) Top 30 most frequent aspect terms.

Figure 2. Exploratory data analysis for the SemEval-2014 laptop dataset.

B ADDITIONAL IMPLEMENTATION DETAILS AND PROMPT TEMPLATES

B.1 TF-IDF + Logistic Regression

For the TF-IDF baseline, each input sentence is annotated with special markers around the target aspect using `[ASP]` and `[/ASP]`. The aspect-marked sentence is then converted into TF-IDF features using unigrams and bigrams, lowercasing, sublinear term-frequency scaling, and a minimum document frequency of 2 to remove rare features. These feature vectors are used as input to a Logistic Regression classifier with balanced class weights and a maximum of 2000 iterations.

Two versions of the model are trained: one on the restaurant training split only, and one on the combined restaurant and laptop training data. To interpret the classifier, we analyse the top feature weights and compute SHAP values over the TF-IDF feature space.

B.2 BERT Base Uncased

We use `bert-base-uncased`, a 12-layer transformer encoder with approximately 110 million parameters, pretrained on English Wikipedia and BookCorpus. Each input is structured as `[CLS] sentence [SEP] aspect term [SEP]`, so that the model jointly encodes the review context and the target aspect. A classification layer is applied on top of the `[CLS]` token to predict the three sentiment classes.

For supervised training, we use weighted cross-entropy loss to account for class imbalance. The model is trained for 6 epochs using the AdamW optimizer, with learning rate 2×10^{-5} , weight decay 0.01, and linear warm-up over the first 10% of training steps. The final checkpoint is selected based on validation macro F1. Experiments are implemented using the Hugging Face Transformers library and run on an NVIDIA T4 GPU.

B.3 Qwen2.5-1.5B-Instruct with LoRA

We adapt Qwen2.5-1.5B-Instruct to three-way aspect-level sentiment classification using supervised fine-tuning with LoRA. Every example uses the same prompt structure during training. The system message is:

```
You classify sentiment toward one target aspect in English reviews.  
Reply with exactly one word: positive, negative, or neutral.
```

The user message is:

```
Review: <review>  
Target aspect: <target>  
What is the sentiment toward this target aspect? Answer with one  
word only.
```

The assistant output is the gold label only: `positive`, `negative`, or `neutral`.

To reduce the effect of class imbalance, we use class-weighted loss based on inverse label frequency in the training split. The laptop-stage LoRA model is initialised from the adapter trained on the restaurant split.

Weighted Loss

The Qwen model is trained with weighted causal language modelling on the templated chat sequence. Cross-entropy is applied only to the assistant tokens corresponding to the polarity label. If the gold token belongs to the target polarity word, its loss contribution is multiplied by the inverse-frequency class weight. The final scalar loss is the weighted average over all valid supervised positions, so that minority classes contribute more strongly during optimisation.

B.4 GPT-5.4-mini Prompting

We evaluate GPT-5.4-mini using both zero-shot and few-shot Chain-of-Thought prompting.

The zero-shot prompt is:

```
Given a sentence and a specific aspect term, classify the sentiment  
expressed toward that aspect as positive, negative, or neutral.
```

The few-shot CoT prompt is:

```
Given a sentence and a specific aspect term, determine the sentiment  
expressed toward that aspect in the sentence. Do this step by step.  
First, locate the aspect term in the sentence. Next, identify the  
words or phrases that are related to the term. The last step is  
to decide if the overall sentiment towards the aspect is positive,  
negative or neutral.
```

This prompt is followed by four manually written examples demonstrating the reasoning process for restaurant and laptop reviews. The model is then applied separately to the restaurant and laptop test sets, and the predicted labels are evaluated using the same metrics as the supervised models.

C SUPPLEMENTARY QUANTITATIVE RESULTS

C.1 Additional Performance Visualisations

Figure 3 provides a visual comparison of per-class F1 scores on the restaurant dataset. Figure 4 and Table 7 summarise the effect of fine-tuning on the laptop dataset.

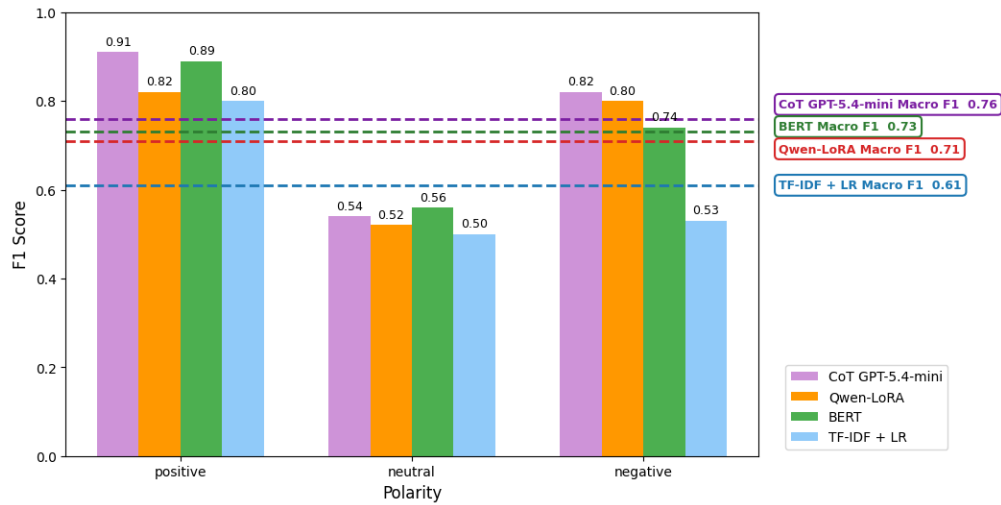


Figure 3. Per-class F1 comparison on the combined restaurant dataset.

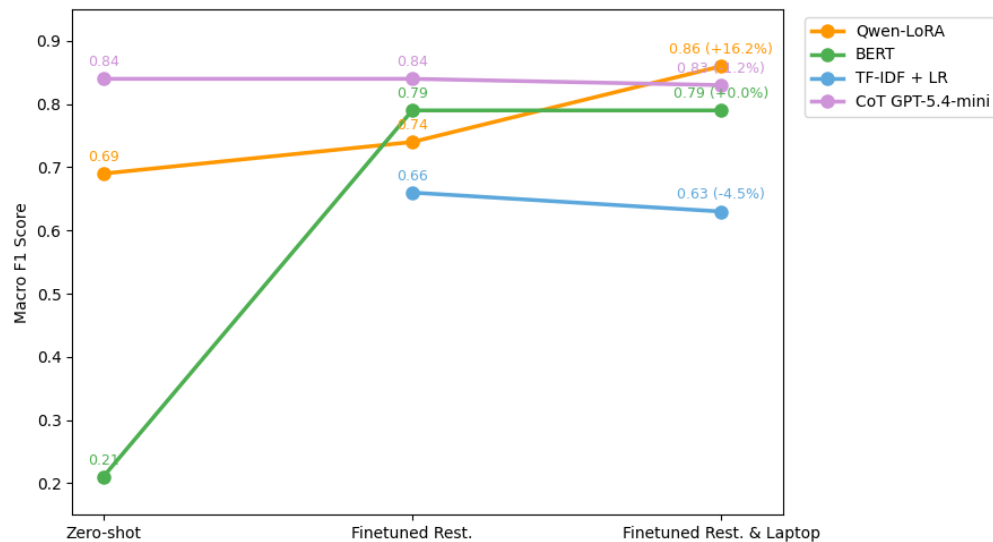


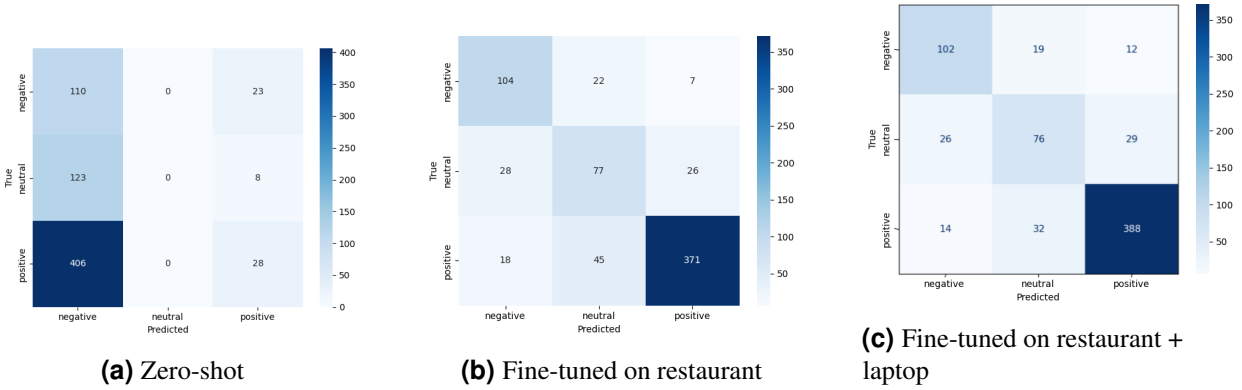
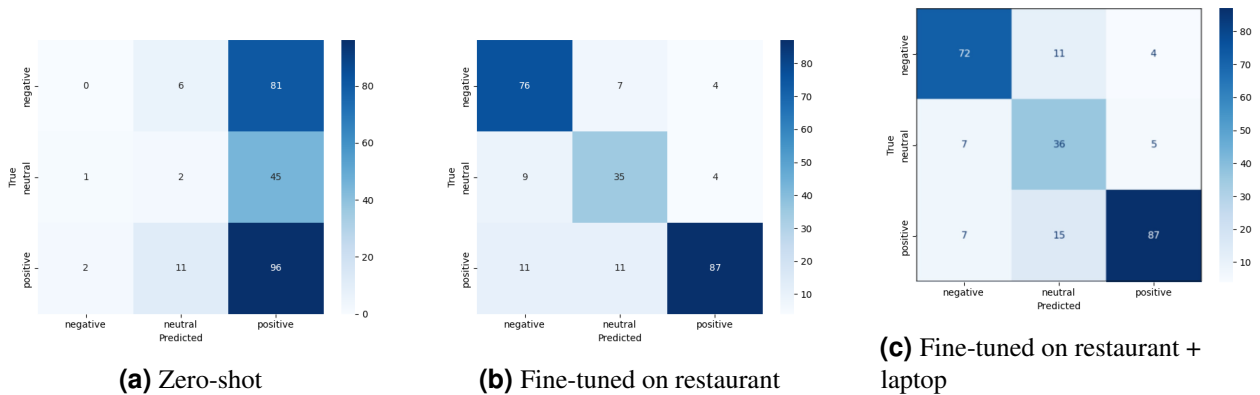
Figure 4. Macro F1 before and after fine-tuning on the laptop dataset.

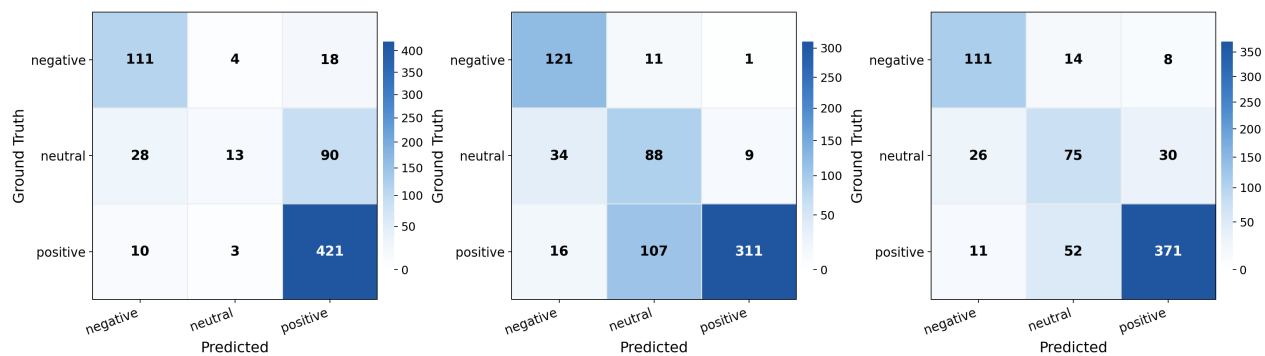
Table 7. Macro F1 on the laptop test set before and after fine-tuning.

Model	Params	Before Fine-tuning	After Fine-tuning	Change (%)
TF-IDF + LR	N/A	0.66	0.63	-4.55
BERT	110M	0.79	0.79	0.00
Qwen-LoRA	1.54B	0.74	0.86	+16.20
GPT-5.4-mini	Undisclosed	0.84	0.83	-1.20

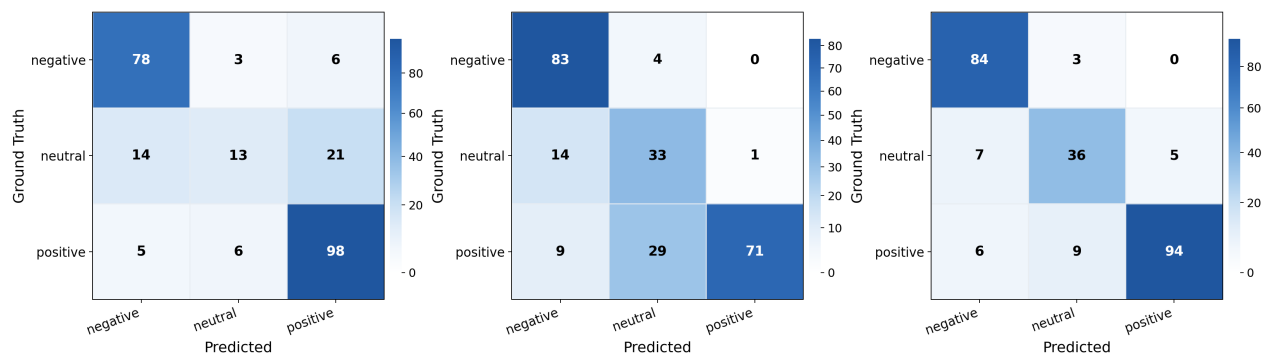
C.2 Supplementary Confusion Matrices

The confusion matrices below provide additional detail on model behaviour across datasets and prompting/training settings.

**Figure 5.** BERT confusion matrices on the restaurant test set.**Figure 6.** BERT confusion matrices on the laptop test set.

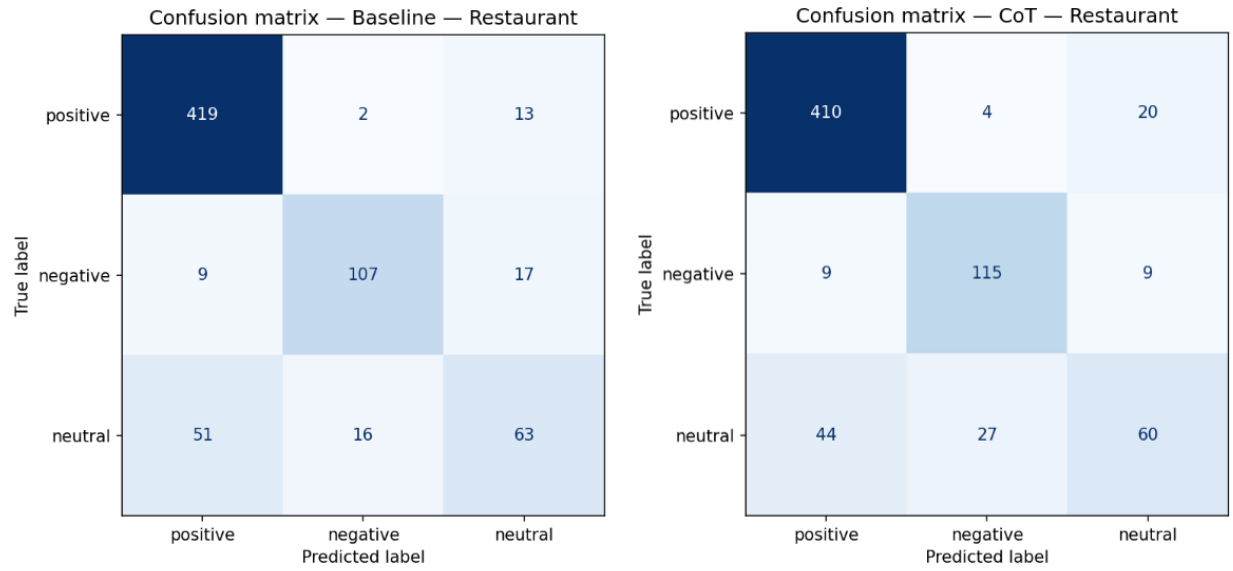


(a) Restaurant test set.

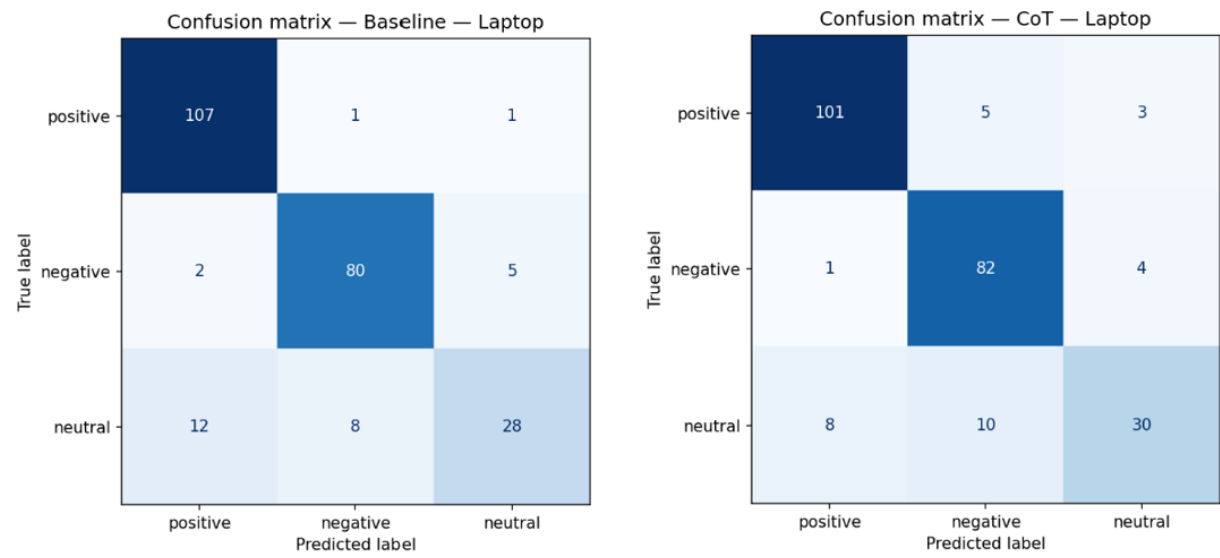


(b) Laptop test set.

Figure 7. Confusion matrices for Qwen: zero-shot, LoRA on restaurant, and LoRA on restaurant + laptop.



(a) Restaurant test set.



(b) Laptop test set.

Figure 8. Confusion matrices for GPT-5.4-mini zero-shot and Chain-of-Thought prompting on the restaurant and laptop test sets.

D SUPPLEMENTARY QUALITATIVE ERROR ANALYSIS

D.1 Restaurant Error Analysis

Table 8 presents representative restaurant-domain misclassification examples across models. The examples show that restaurant errors cluster around a small number of recurring phenomena: implicit sentiment, contrastive sentiment, aspect-level attribution, annotation ambiguity, irony, and discourse-level interpretation.

Table 8. Representative restaurant misclassification examples grouped by error type. Legends: **Green** = positive, **Orange** = neutral, **Red** = negative.

Sentence	ML	BERT	Qwen	GPT	Reason
<i>Implicit sentiment (no explicit polarity cue) — all failed</i>					
Order the Tortilla Soup .	neu	neu	neu	neu	Sentiment is expressed as a recommendation without an explicit evaluative word, leaving no clear polarity cue.
<i>Implicit negative (inference from unmet expectation) — resolved only by Qwen</i>					
Barbecued codfish was gorgeously moist - as if poached - yet the fabulous texture was let down by curiously bland seasoning, however herb mix or other sauce would have done much to enhance.	pos	pos	neg	pos	Negative sentiment is implied through absence; other models are distracted by nearby positive wording.
<i>Contrastive sentiment (polarity scope resolution) — all failed</i>					
The food was very expensive (we spent \$160 for lunch for two) but extremely tasty.	pos	pos	neu	neu	Competing positive and negative clauses require correct sentiment scoping to the target; all models fail to resolve the contrast.
<i>Aspect-level attribution (target-specific sentiment mapping) — all failed</i>					
We didn’t even see a menu, as our waiter described both the specials and the main dishes.	neg	neg	neu	neu	The sentence contains both negative and positive cues, and correct prediction depends on attributing sentiment to the correct aspect.
<i>Neutral boundary (annotation vs linguistic signal mismatch)</i>					
Excellent lunch and thoroughly enjoyed food and ambiance!	pos	pos	pos	pos	Strongly positive wording conflicts with a neutral gold label, reflecting annotation ambiguity.
<i>Irony understatement (local pragmatic inference) — resolved only by GPT</i>					
The only thing the waiters don’t do for you is wipe your chin when you leave.	neg	neg	neg	pos	Irony requires pragmatic interpretation; other models interpret the negation literally.
<i>Discourse-level sentiment (global pragmatic interpretation) — partially resolved</i>					
Very excited to hear that short-term Chef Jason Narone has moved on, he truly was a low point of their Sterling track record.	pos	pos	neg	neg	Negative evaluation is embedded in discourse framing; initial positive cue misleads some models.

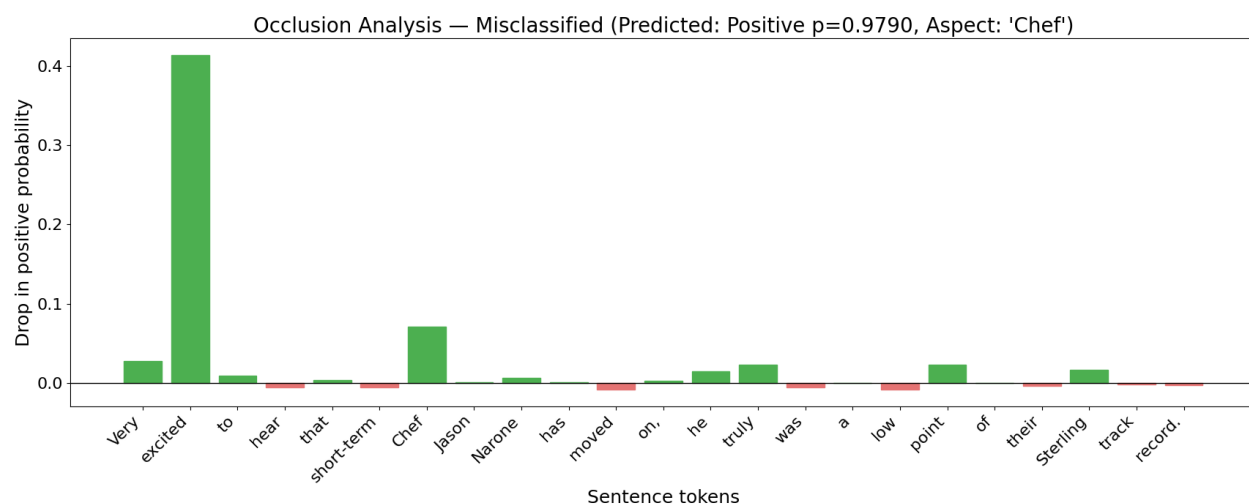


Figure 9. Occlusion-based token importance for BERT on the misclassified “Chef” example from Table 8. Bars show the drop in predicted positive probability when each word is masked.

D.2 Laptop Error Analysis

Table 9 presents representative laptop-domain misclassification examples. Compared with the restaurant domain, these errors more often require technical understanding, conditional reasoning, and interpretation of quantitative comparisons.

Table 9. Representative laptop misclassification examples grouped by error type. Legends: **Green** = positive, **Orange** = neutral, **Red** = negative.

Sentence	ML	BERT	Qwen	GPT	Reason
<i>Aspect-level sentiment attribution (price) — resolved only by GPT</i>					
Disappointing for such a lovely screen and at a reasonable price .	neg	neg	neg	pos	“Disappointing” applies to the screen, not the price; BERT and Qwen misattribute the negative cue to the price aspect.
<i>Numerical comparative reasoning (value retention) — resolved only by GPT</i>					
Reason why? It’s because when you buy it, you know first thing that you will not lose any value for that laptop, the price will stay the same for the next year, and even if Apple does decides to change mode, your laptop value will only drop 10–20%, unlike PC laptops which drop more than 80%.	neg	neg	neg	pos	Positive sentiment is embedded in a numerical comparison; “drop 10–20%” dominates BERT and Qwen despite the overall framing being favourable.
<i>Conditional perfection (negative implied by missing feature) — resolved only by GPT</i>					
It could be a perfect laptop if it would have faster system memory and its radeon 5850 would have DDR5 instead of DDR3.	neu	neu	neu	neg	“Perfect” dominates BERT and Qwen; GPT resolves the conditional structure as implying dissatisfaction with the missing feature.
<i>Double negation resolving to positive — resolved by GPT and Qwen</i>					
It’s graphics aren’t bad at all, for the lower end of the MacBook Pro spectrum, easily capable of running StarCraft II.	neg	neg	pos	pos	“Aren’t bad at all” resolves to positive; BERT attends to “bad” without resolving the double negation.
<i>Technical absence as positive signal — resolved only by GPT</i>					
Drivers updated ok but the BIOS update froze the system up and the computer shut down.	neu	neu	neu	pos	“Updated ok” is a domain-specific positive cue; BERT and Qwen lack exposure to laptop technical vocabulary and default to neutral.
<i>Contrastive optimistic resolution (battery) — all failed</i>					
The battery doesn’t last long but I’m sure an upgrade battery would solve that problem.	neg	neg	neg	neg	The reviewer’s optimistic stance in the second clause expresses positive sentiment toward the battery’s upgradability; all models attend to “doesn’t last long” and ignore the resolution.

E LOCAL SHAP EXPLANATION EXAMPLE

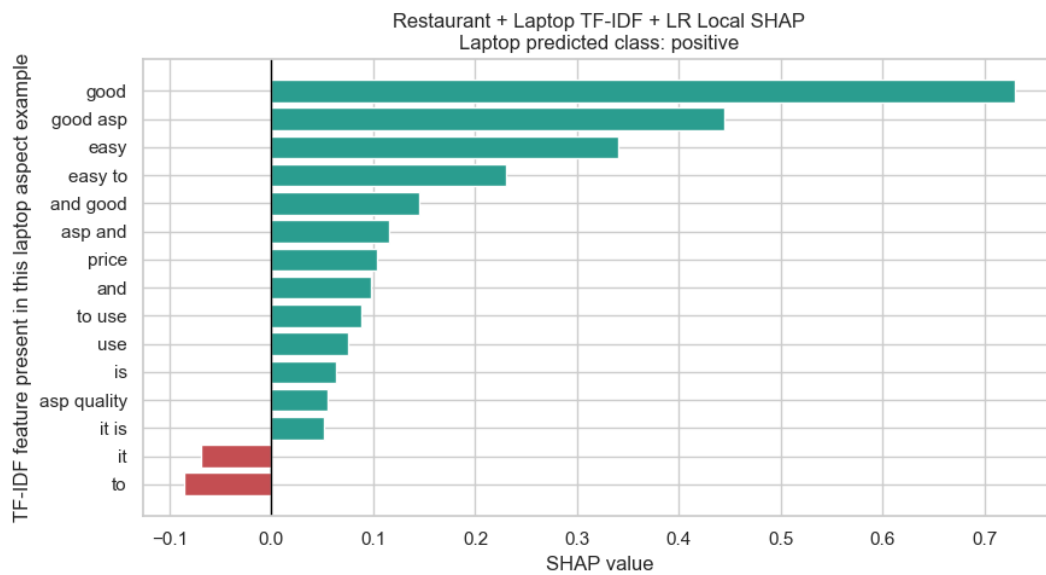


Figure 10. Local SHAP explanation for the sentence *“It is easy to use, good quality and good price.”*, predicted as positive. Positive SHAP values (right) indicate features pushing the prediction toward the positive class (e.g., “good”, “easy”), while negative values (left) indicate features opposing it. The overall prediction is dominated by strong positive sentiment terms, with other tokens contributing only marginally.

F TRAINING HYPERPARAMETERS

Table 10. Qwen LoRA supervised fine-tuning hyperparameters and rationale.

Name	Value	Reason
Base model	Qwen2.5-1.5B-Instruct	Instruction-tuned LM suited to short, constrained outputs.
Adapter	LoRA ($r=16$, $\alpha=32$, dropout 0.05)	Train only adapter weights: cheaper and usually more stable than full-model fine-tuning.
Restaurant	batch 20, accum. 1; RTX 4090	Settings used when training on the restaurant split; sized to GPU memory.
Laptop	batch 6, accum. 2; RTX 4060 Ti	Settings used when training on the laptop split; accumulation partly offsets a smaller per-step batch.
Epochs	3	Enough passes for the task without unnecessary training time.
Learning rate	2×10^{-4}	Typical scale for LoRA fine-tuning.
LR schedule	Cosine, warmup ratio 0.05	Warmup stabilises early steps; cosine decays the rate smoothly.
Max. sequence length	256	Token-length analysis on templated inputs shows short sequences (e.g. train max 198, test max 180); 256 fits the distribution and matches the suggested cap from that analysis.
Class weights	Inverse frequency (3 classes) on train split	Mitigate imbalance by upweighting rarer labels in the loss.

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